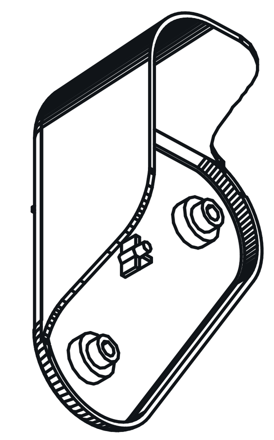
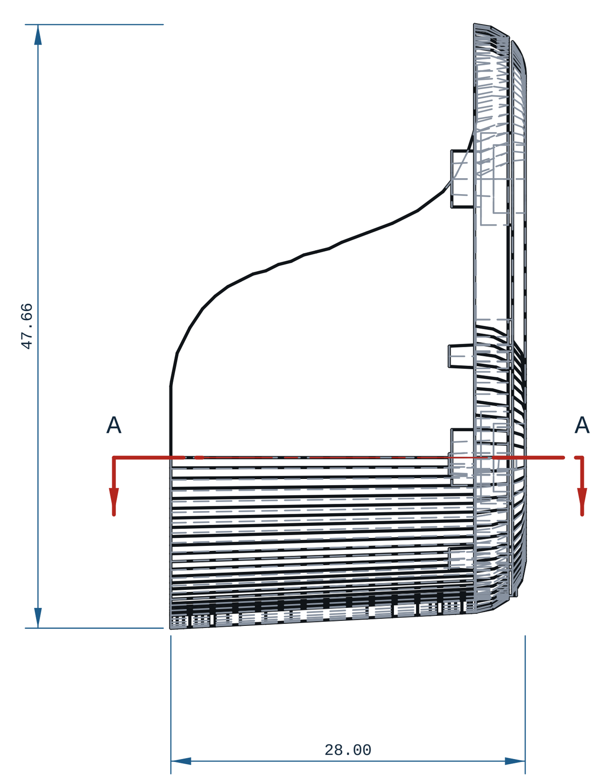
microduck-upper-leg-left   SCALE 2:1   A1   THIRD ANGLE



ISOMETRIC VIEW

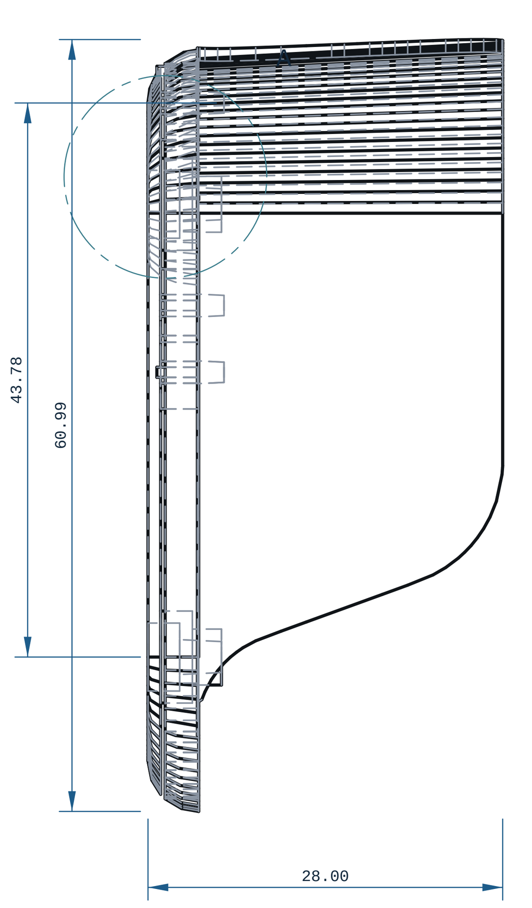
28.00 x 47.66 x 60.99 mm

REFERENCE RENDER (ISO2) ? rendered off this solid, 900 x 680 px



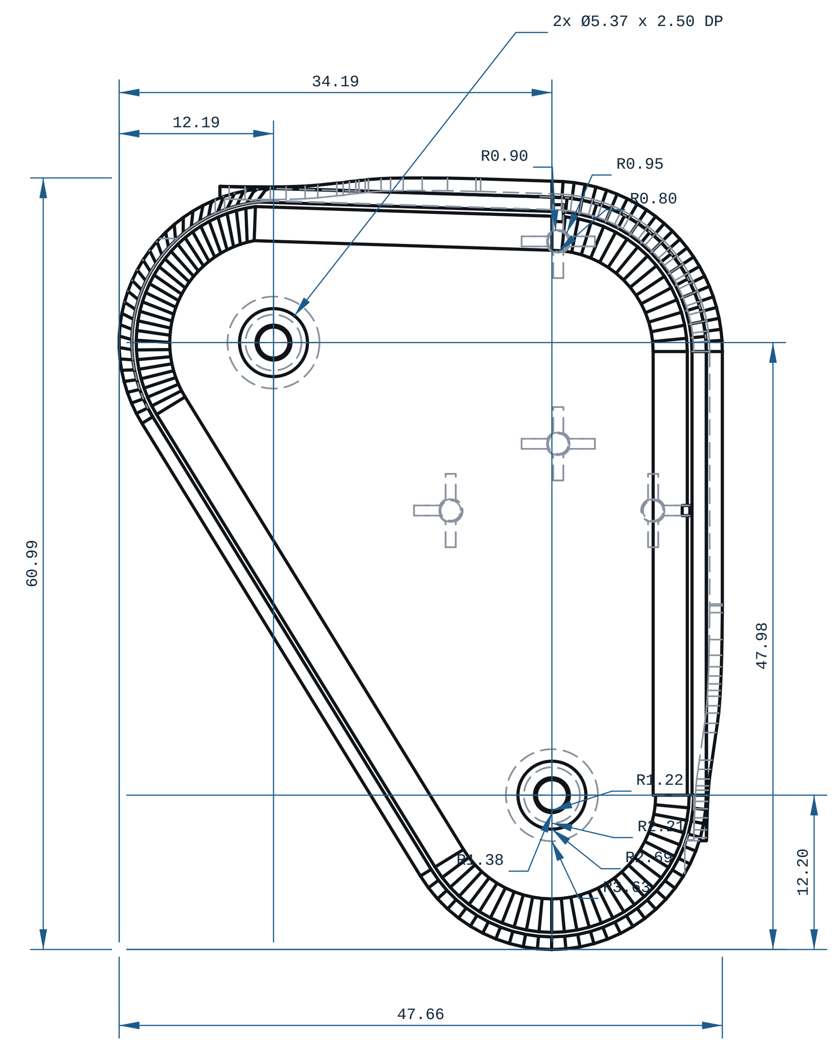
TOP VIEW

X ?   Y ?



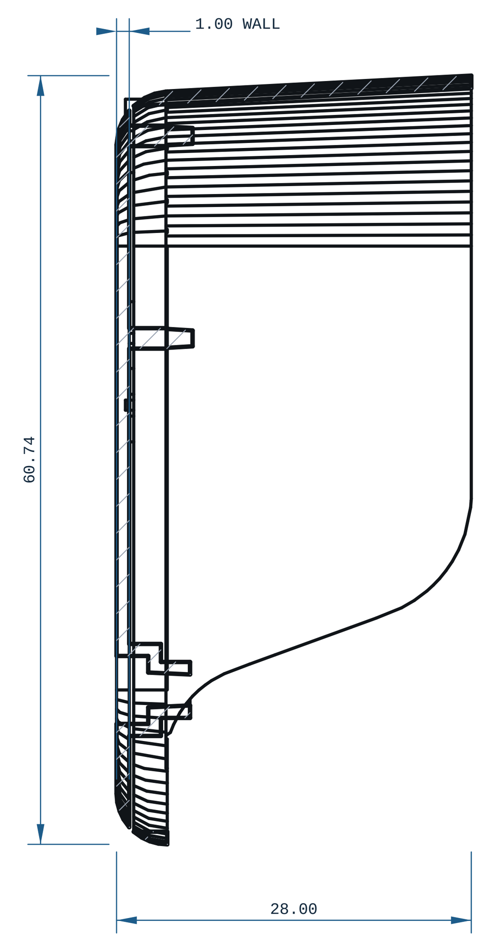
FRONT VIEW

-X ?   Z ?



RIGHT VIEW

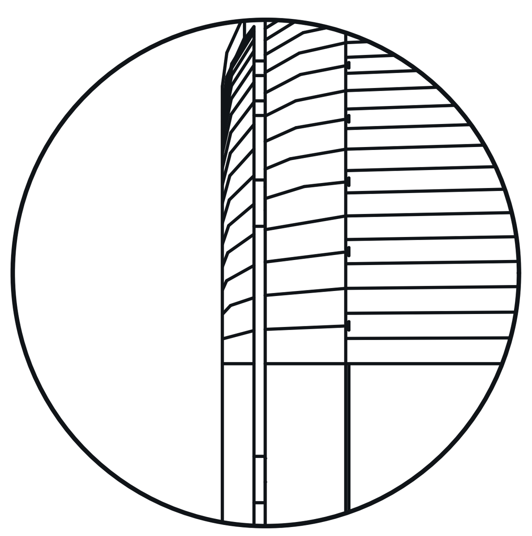
-Y ?   Z ?



SECTION A-A

SCALE 5:1

DETAIL A



NOTES

1. ALL DIMENSIONS IN mm AND DEGREES. EVERY DIMENSION AND EVERY HOLE CALLOUT WAS MEASURED OFF THE SOLID.
2. PROCESS: FDM.
3. THINNEST WALL, WHOLE SOLID (RAY MINIMUM): 0.0240 mm, AT X=69.125 Y=-10.700 Z=37.938 ? MEASURED OVER THE WHOLE SOLID BY RAY+EXACT AT 2.1730 mm SAMPLE SPACING (A FEATURE NARROWER THAN THAT CAN BE MISSED). THIS IS A DISTANCE THROUGH MATERIAL, NOT A WALL: AT AN EDGE, A FILLET RUN-OUT OR A CHAMFER TIP IT IS REAL AND UNPRINTABLE BY DEFINITION. DO NOT JUDGE PRINTABILITY BY IT ? THE NEXT NOTE IS THE ONE THAT DOES.
4. PRINTABLE MINIMUM WALL: 0.7802 mm, AT X=69.542 Y=16.530 Z=6.271 ? MEASURED AS THE SMALLEST MEDIAN WALL OVER A NEIGHBOURHOOD OF 6.5190 mm (>= 8 SAMPLES), RAYS CAPPED AT 4.000 mm, BY cecad.inspect.printable_wall_detail. AGAINST THE 0.8000 mm FLOOR (2 PERIMETERS @ 0.4 mm NOZZLE, cecad/printed.py:801): VERDICT FAIL ? THIS REGION IS BELOW THE FLOOR AND WILL NOT PRINT SOLID: THICKEN IT, OR PRINT WITH A SMALLER NOZZLE AND SAY SO ON THE ROUTE CARD.
5. OUTLINE COMPLEXITY: RIGHT VIEW DRAWS 783 VISIBLE EDGES (COUNTED OFF TechDraw.projectEx; A CLEAN SHOP VIEW ON THIS SHELF RUNS 4-131). THE EXTRA LINES ARE CONSTRUCTION SEAMS OF THE PARAMETRIC REBUILD ? SURFACE PATCH JOINS ? AND ARE NOT FEATURES TO MANUFACTURE. DO NOT WORK TO THEM. WHAT SETTLES IT: REBUILD THE SURFACE AS ONE SMOOTH FACE THROUGH THE SAME MEASURED TABLE INSTEAD OF A STATION-TO-STATION LOFT, THEN REDRAW.
6. GENERAL TOLERANCE CANNOT DETERMINE ? SEE PRINT / DFM NOTES UNLESS STATED.
7. SECTION A-A AT Y=0.00: THINNEST MATERIAL IN THIS PLANE ONLY 1.00 mm, AT X=70.00, Z=-7.64 ? A SCAN OF ONE CUT, NOT THE PART MINIMUM.
8. FRONT VIEW: hidden lines KEPT: without them 10 of 1632 edge samples have no ink within 0.35 mm (worst 2.000 mm) and the feature would be missing from the sheet.
9. TOP VIEW: hidden lines KEPT: without them 2 of 1699 edge samples have no ink within 0.35 mm (worst 0.550 mm) and the feature would be missing from the sheet.
10. RIGHT VIEW: hidden lines KEPT: without them 173 of 1548 edge samples have no ink within 0.35 mm (worst 23.120 mm) and the feature would be missing from the sheet.

PRINT / DFM

1. ORIENTATION: upright, Z-up (28 x 48 x 61 mm)
2. THINNEST WALL, WHOLE SOLID (RAY MINIMUM): 0.0240 mm ? a distance through material, NOT a printable wall where it falls on an edge, a fillet run-out or a chamfer tip. Printability is judged on the next line.
3. PRINTABLE MINIMUM WALL: 0.7802 mm (smallest median wall over a 6.5190 mm neighbourhood, cecad.inspect.printable_wall_detail) against a 0.8000 mm floor (2 perimeters @ 0.4 mm nozzle): VERDICT FAIL ? thicken this region or print it with a smaller nozzle and say so on the route card.
4. 2 horizontal bore(s) bridge at the crown: ream any bearing or press-fit bore to size
5. FILAMENT: PLA, ~5 g (modelled), 305 layers @ 0.2 mm
6. TOLERANCE BASIS, IN-HOUSE: CANNOT DETERMINE. Every Bambu on this farm records tolerance_mm: null, cite "no coupon printed and measured on this machine" (ce-machines/bambu-h2s-*, bambu-h2d-*/capabilities.json, read 2026-09-02), and Bambu Lab's own H2D Pro spec sheet carries no dimensional-accuracy line. DIMENSIONS HERE ARE NOMINAL AS MODELLED.
7. TOLERANCE BASIS, OUTSOURCED: JLC3DP ±0.3 mm FDM (jlc3dp.com, ce-machines/farm-jlc3dp); Hubs ±0.5%, floor ±0.5 mm (hubs.com/3d-printing/, ce-machines/farm-hubs). Apply the route's figure, not both. The ±0.2 mm in cecad/machining.py fdm is a repo planning constant with no vendor cite and is NOT used here. One printed and measured coupon closes this note.
8. NO ISO 286 CLASS IS DECLARED ON ANY BORE: FDM cannot be sized to one (H7/p6 at Ø3 is a 0.010 mm band). REAM bearing and press-fit bores after printing.
9. RADII NAMED IN cad/part.py THAT ARE NOT CIRCULAR EDGES ON THIS SOLID: 4.00 mm, 12.20 mm. cecad.inspect.arc_radii measures 8 ARC(S) HERE AND EVERY ONE IS DIMENSIONED ABOVE. NO DIMENSION IS MISSING: A LEADER NEEDS AN EDGE. THE GEOMETRY FILE CARRIES THE SURFACE.
10. ORTHOGRAPHIC VIEWS: 3 (FRONT, TOP, RIGHT) ? cecad.autosheet.choose_views ON THE 28.000 x 47.661 x 60.991 mm BBOX, THIN/LONG 0.4591. THE FULL THIRD-ANGLE SET IS DRAWN.

MICRODUCK-UPPER-LEG-LEFT

REV A

MATERIAL		PROCESS	SCALE
PLA		FDM	2:1
CATEGORY: housing	GENERAL TOL: CANNOT DETERMINE ?~	FINISH: AS PRINTED	SHEET: A1 1/1
VOLUME (MEASURED): 3.84 cm3		MASS (ASSUMES PLA): 4.8 g	UNITS: mm / deg
SIZE (BBOX, MEASURED): 28.00 x 47.66 x 60.99		DRAWN: ce-cad 2026-09-03	PROJECTION: THIRD ANGLE
SOURCE (DESIGN FILE): ce-parts/microduck-upper-leg-left/current/cad/part.py			